



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

November 27, 2025

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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Frequentist View

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Continuous Distributions

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■ *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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■ *Gaussian Distribution:* $N(\mu, \sigma) = f(y|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{y-\mu}{\sigma}\right)^2\right)$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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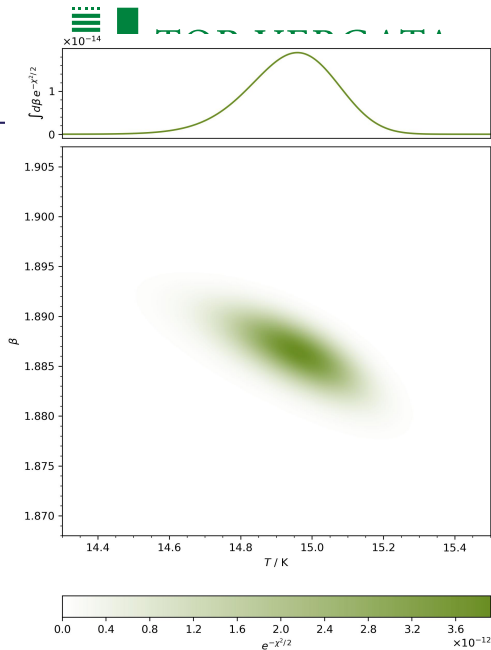
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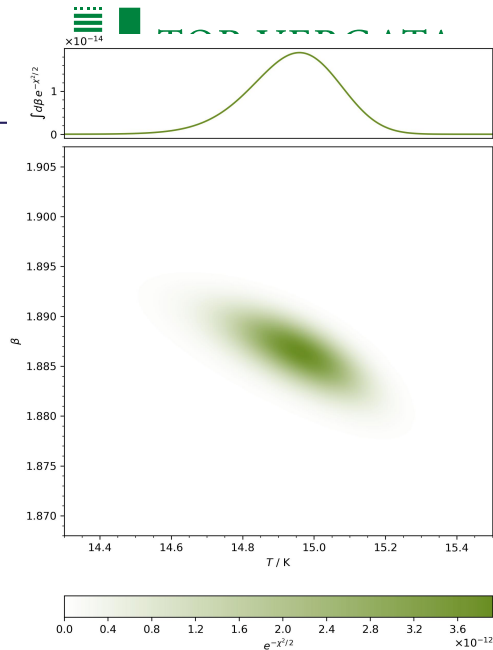
Alternatives

Where I Studied

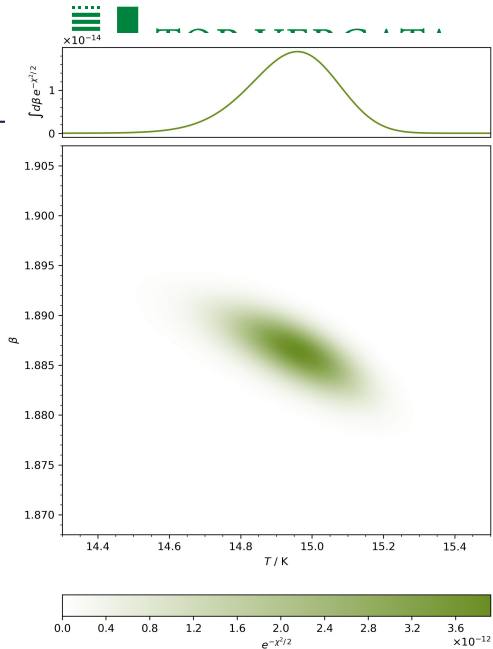
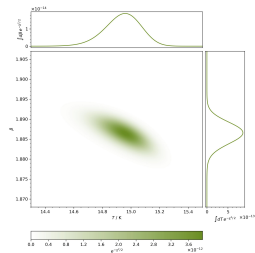
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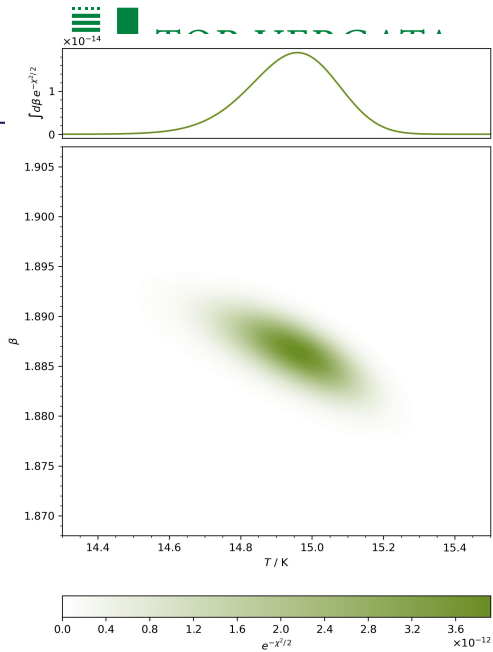
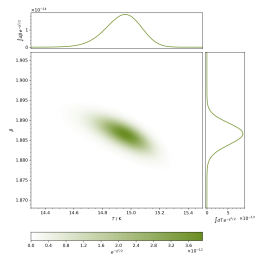
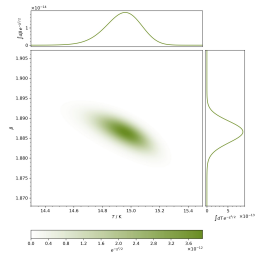
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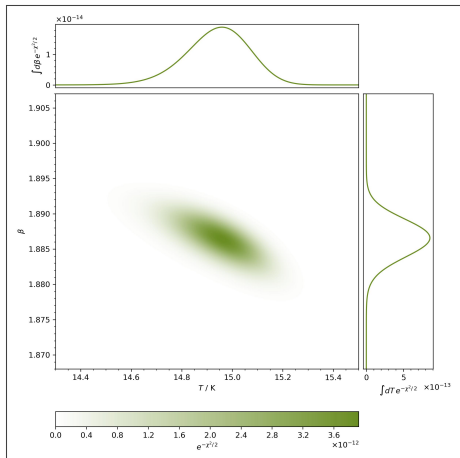


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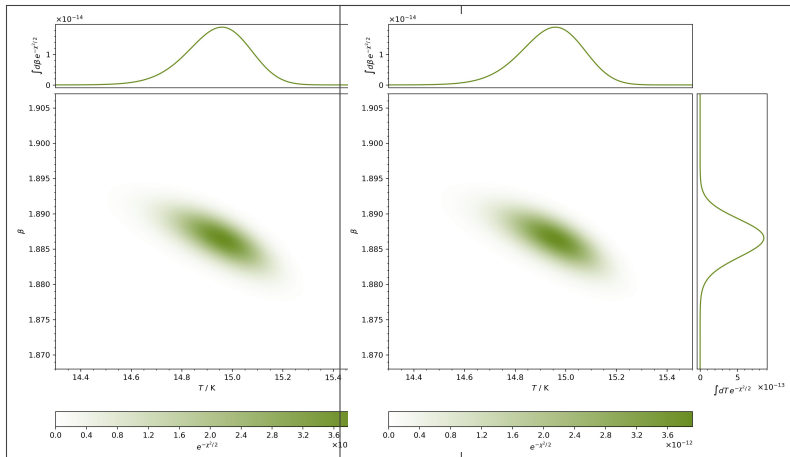


What I Studied

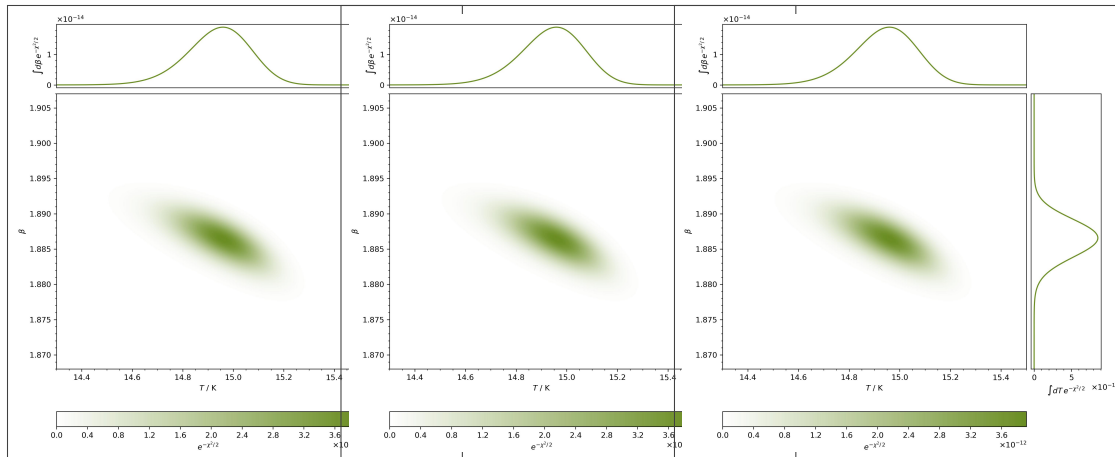
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Why Astrophysics

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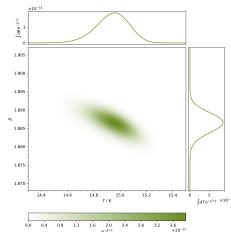
- Interesting

Why Astrophysics

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 - Awesome Scales

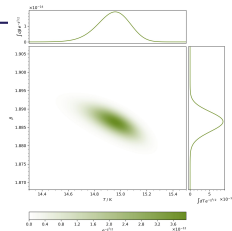
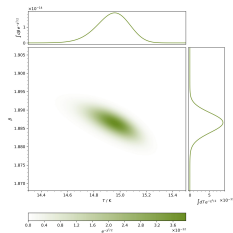
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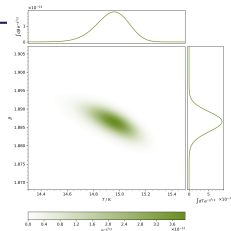
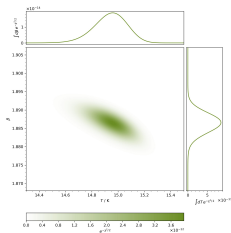
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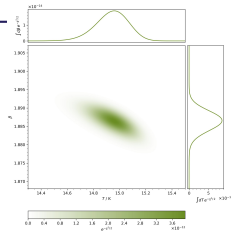
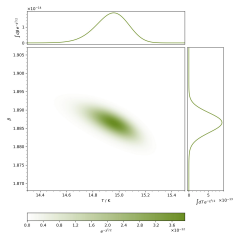
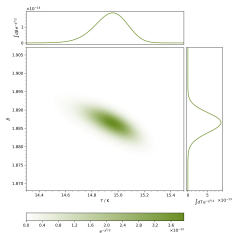
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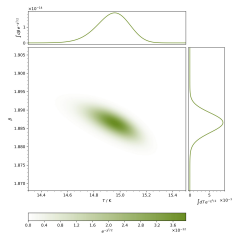
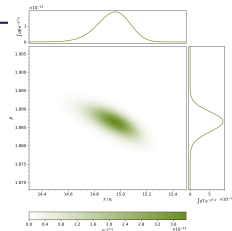
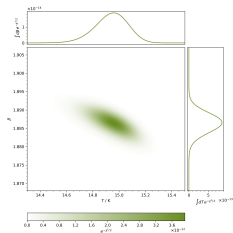
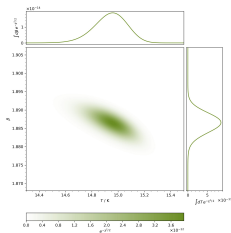
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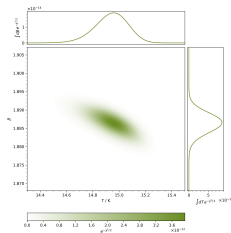
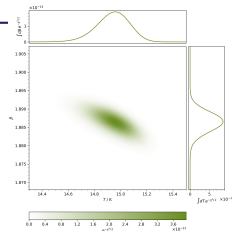
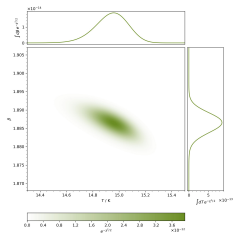
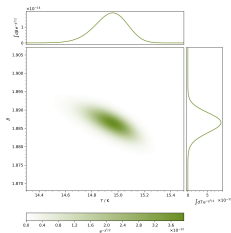
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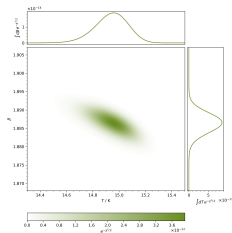
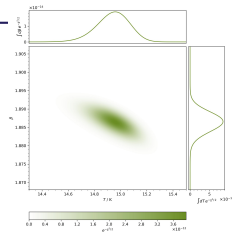
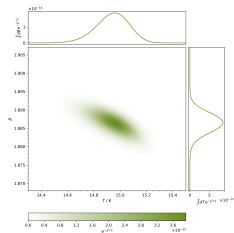
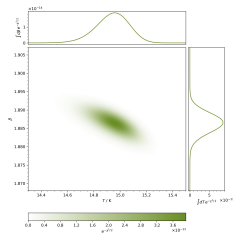
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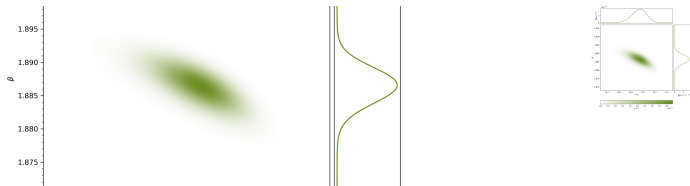
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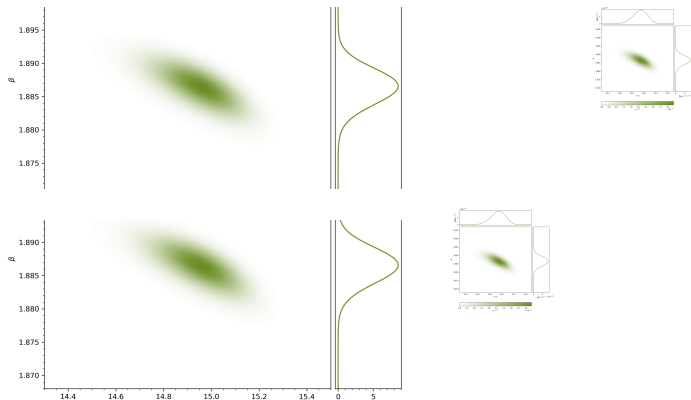
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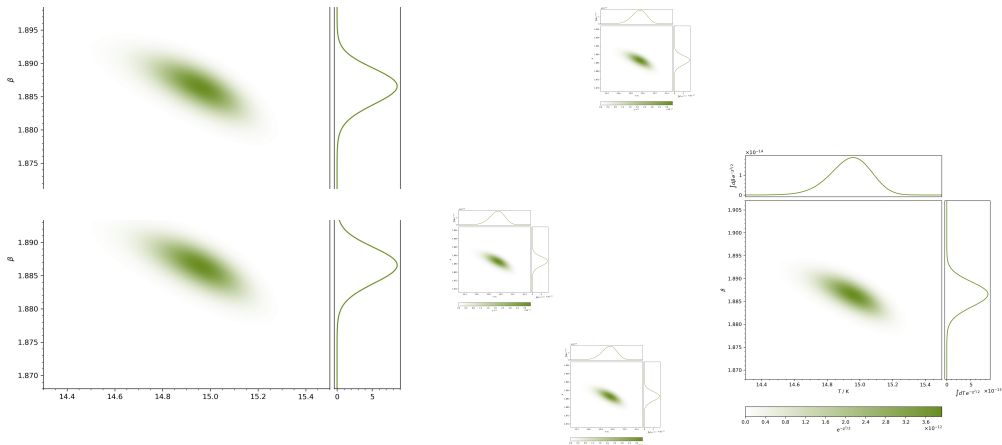
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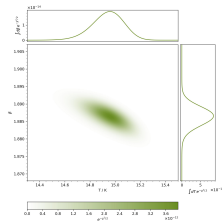
Other Interests

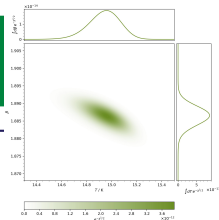
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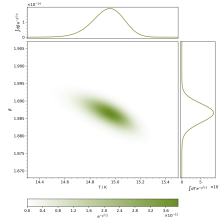
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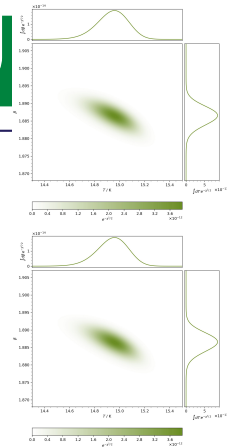
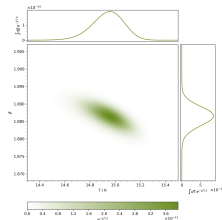
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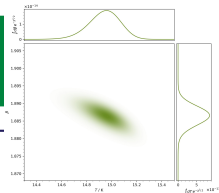
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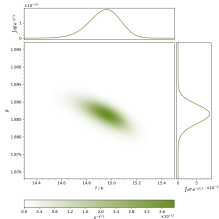
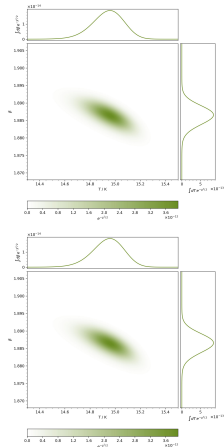
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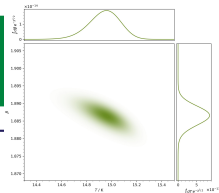




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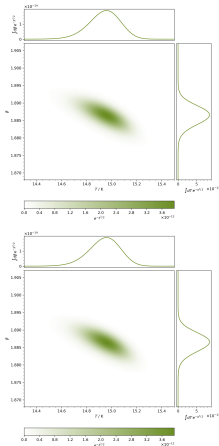
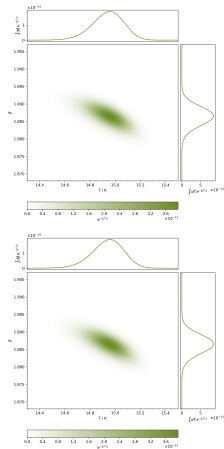
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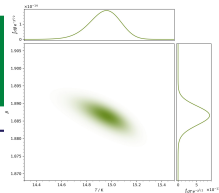


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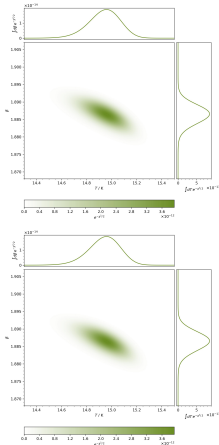
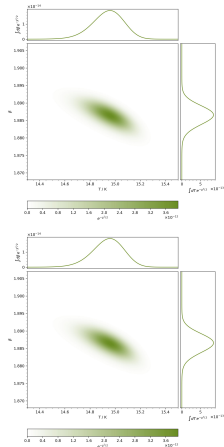


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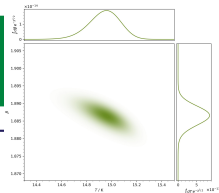


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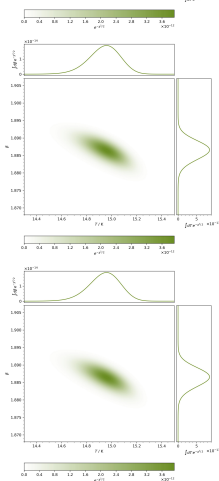
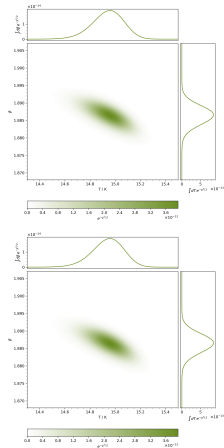


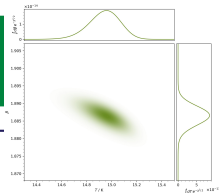
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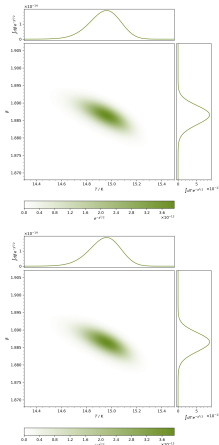
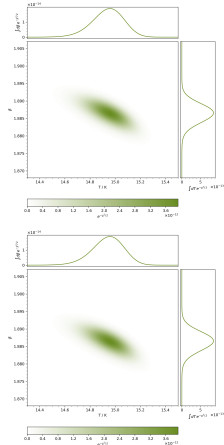
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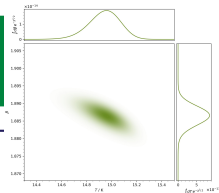




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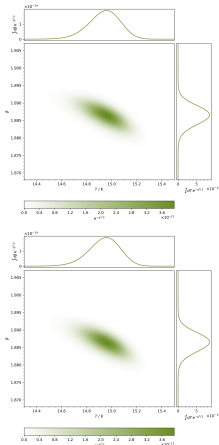
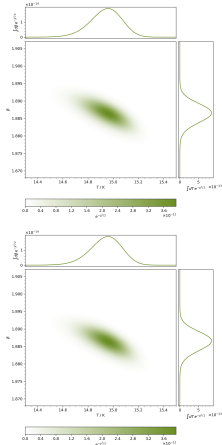
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Other Interests

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- Programming
- Politics
- Meeting New People



Optimisation

Thank You!